

# MUHIL AMUTHAN M

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## Professional Summary

Pre-final-year Electronics & Communication Engineering student (Minor: Computer Science, graduating 2028) with hands-on experience in full-stack development, applied AI/ML, and IoT/embedded systems across live internships and competitive hackathons. Ranked top 25 of 3,700+ teams at Quest Global Ingenium 2026; built and shipped 4+ end-to-end ML and full-stack projects. Proficient in Python, Java, FastAPI, React, and embedded systems; solved 80+ LeetCode problems.

## Education

### V.S.B. Engineering College, Karur

Expected 2028

*B.E. Electronics & Communication Engineering (Minor: Computer Science)*

*Tamil Nadu, India*

- CGPA: 7.94/10
- Coursework: Data Structures & Algorithms, OOP (Java), Computer Networks, Microcontrollers, Machine Learning

## Experience

### Techzy IT Solutions

Jun 2026

*AI/ML Intern*

*Coimbatore, Tamil Nadu*

- Built and deployed a machine learning delivery-time prediction application, iterating through version control on GitHub to ship a working live demo.

### BSNL India, Madurai

2025

*Implant Training Intern*

*Madurai, Tamil Nadu*

- Gained hands-on exposure to telecom infrastructure, switching systems, and real-time OSI-layer communication.

## Projects

### CarbonWise — Real-Time Grid Carbon Intelligence | *Python, FastAPI, React, ML, Cloud DB* Jan – Mar 2026

- Ranked top 25 of 3,700+ teams at Quest Global Ingenium 2026; presented at India AI Impact Buildathon 2026.
- Built an end-to-end ML platform computing grid carbon intensity at 15-minute intervals, surfacing the top-5 low-CO<sub>2</sub> windows via a FastAPI backend and React/Recharts dashboard (41 commits).

### Smart PPE Compliance Monitoring & Interlock System | *ESP32, ESP-NOW, RC522 RFID, Embedded C/C++* 2025 – 2026

- Engineered an IoT industrial-safety system that authenticates workers via RFID and verifies PPE compliance (helmet, safety vest) before enabling machine operation.
- Implemented ESP32-to-ESP32 wireless communication over ESP-NOW with relay-based machine interlock and LED/buzzer alerts for low-cost, real-time safety enforcement.

### Food Delivery Time Prediction | *Python, Scikit-learn, Streamlit*

2026

- Built and benchmarked Linear Regression and Decision Tree models predicting delivery time from distance, weather, traffic, and courier experience.
- Deployed as an interactive Streamlit app for real-time predictions.

## Technical Skills

**Languages:** Python, Java, SQL, JavaScript, HTML/CSS, Embedded C/C++

**Frameworks & Tools:** FastAPI, Flask, React, Streamlit, Git/GitHub, REST APIs, VS Code, IntelliJ IDEA

**AI/ML:** Machine Learning, Deep Learning, CNN, NLP, Generative AI, Federated Learning, Differential Privacy, Scikit-learn

**CS Fundamentals:** Data Structures & Algorithms, OOP, System Design, Computer Networks (OSI/UDP)

**Electronics/IoT:** Microcontrollers, IoT Architecture, Analog Circuits, Embedded Systems

## Certifications

Fundamentals of Generative AI (AWS Training & Certification) • Deep Learning & Generative AI (Infosys Springboard) • Intro to IoT (Cisco) • IoT & Digital Transformation — NASSCOM FutureSkills Prime (Gold, 74%) • Analog Circuits — NPTEL, IIT Kanpur

## Achievements

Top 25 of 3,700+ teams, Quest Global Ingenium 2026 • India AI Impact Buildathon 2026 Participant • 80+ LeetCode Problems (50-Day Badge) • 12 Public GitHub Repositories • CRYPTERA 2026 Presenter